ABHISHEK TIWARI

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PROFESSIONAL SUMMARY

Computer Engineering student at PSIT Kanpur (2026) building full-stack applications and ML systems using Python, React, and Node.js. Delivered a 95%-accurate Fake News Detector and a Netflix-style Recommendation Engine with collaborative filtering. Certified by Harvard (CS50) and Stanford (Machine Learning Specialization)

SKILLS

- **Programming Languages:** Python, JavaScript
- **Frameworks & Libraries:** React, Node.js, Express
- **Tools:** Git, IntelliJ, Jupyter Notebook, VS Code
- **Databases:** MySQL, PostgreSQL
- **Methodologies:** Agile, Scrum, SDLC
- **Specializations:** Full-Stack, Backend, Frontend

PROJECTS

Netflix Recommendation System

Python · TensorFlow · REST API · React · Flask

- Cleaned and pre-processed a dataset of over 17,000 Netflix titles, addressing missing values across 10+ feature columns to ensure data quality.
- Developed a TF-IDF content similarity engine, reducing recommendation latency to 200ms and delivering personalized results across more than 5,000 titles.
- Implemented collaborative filtering techniques, achieving 85% precision in predicted user ratings during offline evaluations to enhance user experience.
- Delivered an intuitive React frontend featuring over 3 UI components, successfully integrating 2 machine learning models through REST API calls for optimal performance.

Fake News Detection

Python · TensorFlow · Scikit-learn

- Trained a classification model on 10,000+ news articles, achieving 95% accuracy on validation sets
- Implemented NLP preprocessing techniques such as tokenization, stopword removal, and TF-IDF
- Reduced feature dimensionality by 60% through effective data preprocessing
- Engineered 15+ insightful text-based features, including sentiment score, word count, and n-gram frequency

Fabric Classification System

Python · TensorFlow · Keras · MobileNet · OpenCV · Streamlit · Grad-CAM · Matplotlib

- Developed a lightweight CNN-based fabric classification system using a modified MobileNet architecture to differentiate between cotton and blended fabrics, achieving an impressive 85.71% test accuracy.
- Optimized the model for deployment, reducing architecture size to 1.09M parameters (~5MB) while preserving performance levels comparable to larger models such as ResNet50 and VGG16.
- Constructed a comprehensive preprocessing and augmentation pipeline utilizing Keras ImageDataGenerator, effectively expanding the training dataset by 63× to enhance model generalization capabilities.
- Created a Streamlit web application featuring real-time predictions, confidence visualization, and Grad-CAM explainability heatmaps, aiming to support interpretable AI-based classification.

EDUCATION

Bachelors of Technology, Computer Engineering
Pranveer Singh Insitute of Technology
December **2022 - August 2026 | Kanpur, India**
CGPA: 7.1

CERTIFICATIONS

- CS50 - Harvard University (edX)
- Machine Learning Specialization - Stanford University
- SQL - DataCamp